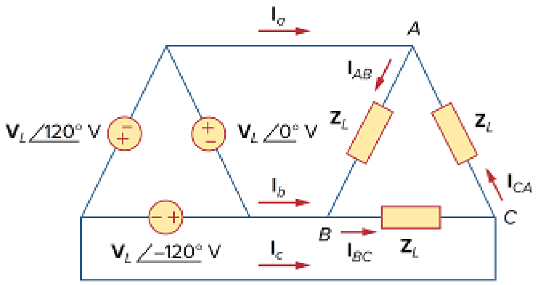
Chapter 12, Problem 20P

Problem

Using Fig. 12.51, design a problem to help other students better understand balanced delta-delta connected circuits.

Figure 12.51



Step-by-step solution

Step 1 of 2

From the given circuit,

The phase currents are,

$$I_{AB} = \frac{V_L\angle 0^\circ}{Z_L}$$

$$I_{BC} = \frac{V_L\angle -120^\circ}{Z_L}$$

$$I_{CA} = \frac{V_L\angle 120^\circ}{Z_L}$$

Step 2 of 2

And the line currents are,

Applying KCL at node *A*,

$$I_a = I_{AB} - I_{CA}$$

$$I_a = \left(\frac{V_L\angle 0^\circ}{Z_L}\right) - \left(\frac{V_L\angle 120^\circ}{Z_L}\right)$$

$$\therefore I_a = \frac{(V_L\angle 0^\circ) - (V_L\angle 120^\circ)}{Z_L}$$

Applying KCL at node *B*,

$$I_b = I_{BC} - I_{AB}$$

$$I_b = \left(\frac{V_L\angle -120^\circ}{Z_L}\right) - \left(\frac{V_L\angle 0^\circ}{Z_L}\right)$$

$$\therefore I_b = \frac{(V_L\angle -120^\circ) - (V_L\angle 0^\circ)}{Z_L}$$

Applying KCL at node *C*,

$$I_C = I_{CA} - I_{BC}$$

$$I_C = \left(\frac{V_L\angle 120^\circ}{Z_L}\right) - \left(\frac{V_L\angle -120^\circ}{Z_L}\right)$$

$$\therefore I_c = \frac{(V_L\angle 120^\circ) - (V_L\angle -120^\circ)}{Z_L}$$